

ASTR 530 115 Homework 1

Due date: 6th October 2026, no later than 14:20.

General Instructions

For this homework, you are asked to write a short report that addresses all of the requirements and questions below. You are free to arrange the information however you like, the order is not very important. Please upload the report to eLearn in **PDF format** (*not* MS word or similar).

Please do not provide more than 4 A4 pages, including pictures. 2 A4 pages is probably enough for most people.

This homework is due in **two** weeks, so you have one class to ask in detail about any problems or questions. Make good use of that opportunity!

You may use LLMs (AI) to help. I **strongly** suggest you try to do things without AI first and then use it to check your answers. Evidence of copying AI output without critical thinking may lower your grade. You should be able to explain what you've written, if asked.

Please see the homework grading information and suggestions on e-Learn.

Useful Links

- NASA Extragalactic Database: <https://ned.ipac.caltech.edu/>
- DESI Legacy Imaging Survey Viewer: <https://www.legacysurvey.org/viewer>
- NASA-Sloan large galaxy atlas: <https://nsatlas.org/>
- Siena Galaxy Atlas: <https://sga.legacysurvey.org/> (for useful background reading, see also the paper: <https://ui.adsabs.harvard.edu/abs/arXiv:2307.04888>)

Specific Instructions

The aim of the homework is to find, interpret and report information about one particular nearby galaxy of your choice. You will need to do some investigation for yourself.

Step 1. Find a galaxy that satisfies the following criteria:

- Appears in the DESI Legacy Imaging Survey;
 - Apparent size large enough to see at least some resolved stars in the Legacy survey image;
 - Has a redshift measurement in NED;
 - Morphologically late type;
 - Not one of the galaxies shown as examples in the lecture slides.
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- Provide the catalog name for your galaxy, and its sky coordinates in decimal degrees.

Step 2. Report the following things about your chosen galaxy:

- Provide an optical image of your galaxy from the “DR11” view, and an near-IR picture from the “unWISE W1/W2 NEO6” (for example, by taking a screenshot). Try to match (and indicate) the angular scale of both images).

Hint: notice that you can adjust the brightness and contrast using the sliders in the lower left.

Hint: the optical image sometimes looks bad for very large galaxies; you may switch the SDSS view instead in this case.

- On one of your images, label any structural features that you can identify.
- Calculate the Hubble flow distance to the galaxy based on its redshift.
- Report at least one redshift independent distance to the galaxy. You may find there is more than one distance in the literature: if so, choose the most recent or use the mean distance from NED.

You may find there is no redshift-independent distance in the literature. If so, comment on which method might be suitable for obtaining such a distance.

- Find a measurement of angular size for the galaxy.

Hint: look for the SGA overlay option in the Legacy Survey viewer.

Step 3. Find and/or compute the following quantities:

- Report an apparent magnitude, absolute magnitude, total luminosity size for your galaxy, in two different filters.

Comment on these filters: what region of the spectrum do they cover? Are the magnitudes in the Vega or AB system? Report the magnitude of the Sun in the same filters (or close equivalents).

If you find some or all of these values in the literature (e.g. in an online database), comment on what fraction of the galaxy light they include.

If you cannot find all these values online (for example, the absolute magnitude but not the apparent magnitude or luminosity), use the equations given in the lectures to derive the missing values.

If you can find all the values online, use the equations to demonstrate they are consistent with each other. If there is any inconsistency, try to explain it.

Hint: remember that you have already found the distance to the galaxy in step 2.

- Report the color of the galaxy using these two filters. How does your result compare to the color of the Sun in the same filters?
- Using the information you have gathered above, and making simple but reasonable assumptions about the surface brightness profile of the galaxy, estimate the surface luminosity density at the centre of the galaxy. Also estimate the effective radius, if you did not find it above.
- Compare your values for the magnitude and effective radius of the galaxy to the size-mass relation graph from Shen et al. (2005) shown in Lecture 3. Comment on what you find.